

CHINA PROPERTY

Expert call series: Ongoing structural divergence along the recovery path

On Jun 2, we hosted the founder of Trend Animals, Mr. Nie Cong (聂聪) to kick off our China Property Expert Call series in 2026. Trend Animals is a digital analysis platform that focuses on China's real estate market cycles and trend-trading strategies, leveraging proprietary indices compiled from high-frequency, city-level secondary property data (across 80+ cities and classified with different styles based on unit sizes, prices, product positionings, locations, etc.).

Overall, the expert observed that the nationwide secondary housing **price decline moderated entering May-26** and highlighted two notable trends: **1)** structural segment out-performance in Shanghai (led by small-size and low-total-price units, likely driven by government-led buybacks for the use of affordable rental housing) and Shenzhen (led by housing units catering to upgrade demand); **2)** price bottoming-out for high-rental-yield assets (incl. residential projects in Urumqi, tourism/vacation/retirement homes in cities of Yunnan, certain low-price housing in Chengdu, etc.).

Looking ahead, the expert sees two paths to recovery: Scenario 1 (the expert's base case) - prolonged structural divergence with price/volume recoveries from new-generation, superior-quality housing and high-yield, low-price assets gradually spilling over to mid-tier housing at different paces, depending on inventory scale (inventory month in leading T&2 cities ranged from 9~32 mths as of Apr-26, vs. 14~479 mths for rest of T2 cities and even higher for T3s); Scenario 2 - broad-based recovery similar to HK, under large-scale policy stimulus. For key market's **recovery timeline** (i.e. Shanghai), the expert expects another 6 months may be needed for price to stabilize under an optimistic case, or 6~12 months under a pessimistic case (vs. our prior discussion on SH/SZ potentially bottoming out in late 2026).

In addition, the expert noted **1) housing consumption** is shifting from location-centric toward more quality-focused (e.g. property management services, community facilities), and is already starting to manifest in secondary prices; and **2)** strong demand trends by lifestyle-driven vacation hubs for diverse buyer cohorts. From an **investment perspective**, in a broader-based recovery scenario, the expert believes key opportunities lie in **1)** high-quality new residential projects, **2)** oversold mid-tier assets in major cities, and **3)** prevailing high-rental-yield properties positioned for value accretion under rate cuts.

Yi Wang, CFA
+86(21)2401-8930 |
yi.wang@goldmansachs.cn
Goldman Sachs (China) Securities
Company Limited

Shi Xu
+86(21)2401-8929 |
shi.xu@goldmansachs.cn
Goldman Sachs (China) Securities
Company Limited

Kaiyan Jing
+86(21)2411-8092 |
kaiyan.jing@goldmansachs.cn
Goldman Sachs (China) Securities
Company Limited

Goldman Sachs does and seeks to do business with companies covered in its research reports. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of this report. Investors should consider this report as only a single factor in making their investment decision. For Reg AC certification and other important disclosures, see the Disclosure Appendix, or go to www.gs.com/research/hedge.html. Analysts employed by non-US affiliates are not registered/qualified as research analysts with FINRA in the U.S.

Related research

China Property: Positioning ahead (No.2): Share rally - what's priced in

China Property: Positioning ahead of Tier-1 cities turnaround

China Property: 2026 Outlook: New uncertainties from continued weak housing market

Key takeaways

1. Differentiated trends captured by Trend Animals: According to the expert, Trend Animals' proprietary housing indices have **led** mainstream public sentiment and official industry statistics by **1-3 months**, utilizing noise-reduction algorithms to identify market inflection points. YTD on a nationwide scale, the expert sees positives in **1)** nationwide price index has seen narrower monthly price declines capped at -0.5% mom after early-year pressure, and has marginally improved to a pace of -0.4% mom in recent weeks, halting the sharp downturn seen in 2H25 despite historically adverse post-Q1 seasonality; **2)** National rental index for 80 cities bottomed in Jan-26, ending a prolonged decline since 2023 with Shenzhen at the forefront of the rental recovery. The expert also highlighted on a by-city level:

- **Structural out-performance in SH/SZ**, (aligning with our deep-dive analysis) whereby for Shanghai, micro-sized units (under 50 sqm) and small-sized units (50-70 sqm) have climbed 4%/11% since bottoming in Nov-25/Jan-26, respectively, as supported by Shanghai government's accelerated existing home buyback initiatives. Meanwhile, Shanghai's commercial apartments (some priced at only 20%-25% of adjacent residential property prices and featuring compelling rental yields) have secured four consecutive months of price appreciation; for Shenzhen, overall housing prices have stabilized over the past 3 months, with mid-to-high-end homes priced between Rmb10mn-20mn having bottomed and edging higher amid recovering upgrade demand. Core sub-markets including Nanshan and Futian have registered clear upward momentum benefiting from resilient upgrade demand.
- **Price bottoming-out for high-yield non-core cities:** Examples include **1) Urumqi:** Select core areas are seeing price increases supported by a 3.6% average rental yield, halting price declines seen for six consecutive months; **2) Dali:** Select Tourism- and retirement-oriented projects rose 8% over the past year, underpinned by ~3% rental yields and high transaction volume.
- **Majority of key cities still sluggish:** BJ's/GZ's home prices remain broadly pressured across segments, though Guangzhou's older small units are showing modest improvement on the back of recently announced local government buybacks. The expert also noted that high-end assets in core districts of Hangzhou and Chengdu have seen accelerated price corrections since Mar-26, though Chengdu's older small-size units outperformed supported by average rental yield of 2.7% (with select units yielding above 3.5%) and increasing allocation from non-local investors.

2. Looking ahead - paths to recovery and timeline

- **Scenario 1** - the expert's base case, no large-scale property stimulus: Prolonged structural divergence with price/volume recoveries from new-generation, superior-quality housing and high-yield, low-price assets gradually spilling-over to mid-tier housing via replacement demand transmission. The expert sees this occurring at different paces, depending on inventory scale, e.g. inventory month in leading T1&2 cities ranged from 9~32 mths as of Apr-26, vs. 14~479 mths for rest of T2 cities and even higher for lower-tier cities. **Scenario 2** - massive supportive

policies (e.g. 50-100bp mortgage rate cuts, significantly accelerated government-led inventory buyback, etc.): the expert expects a recovery pattern to follow HK, with quick expansion in risk appetite and fast price rebound in mid-price-range housing in key cities.

- **What could be Shanghai's recovery timeline?** The expert expects another 6 months may be needed for price to stabilize under an optimistic case, or 6-12 months under a pessimistic case (vs. our prior discussion on SH/SZ potentially bottoming out in late 2026). In select districts (e.g. Putuo), the expert noted already observing meaningful transmission from active “Lao Po Xiao (old, dilapidated, and small)” home transactions to volume uptick for housing units above 100sqm. On the other hand, digestion for housing units sized between 50~90 sqm could be more strenuous due to higher inventory under the prior “7090” rule (effective in 2006 to 2024).

3. Housing demand trends from consumption/investment perspectives

- **“New Consumption” trends:** the expert noted that market's core pricing rationale has been shifting from a traditional “location-centric” focus toward increasing weighting on property services and living qualities, as showcased by **1)** 15%-20% unit price premium for projects offering superior property management services and enhanced facilities (clubhouse, pedestrian-vehicle separation, and elevator retrofitting etc.) relative to adjacent peer projects; **2)** lifestyle-driven demand for high-quality vacation/retirement housing in key tourism hubs (e.g. Dali, Tengchong, Pu'er, and Beihai), underpinned by divergent buyer cohorts ranging from retirees seeking post-retirement living solutions, to remote-working young professionals prioritizing lifestyle flexibility.
- **Investment** - The expert views the following property categories as attractive investment targets amid a more broader-based recovery trajectory: **1)** new-generation, high-quality residential projects with strong price appreciation potential once circulated in secondary markets; **2)** oversold mid-tier assets in key cities such as Suzhou/Nanjing/Hangzhou/Wuhan since this downturn, which might stage sharp rebounds similar to HK's recent mid-tier-home rally; **3)** high-rental-yield properties (prevailing >3.5%) might see 20-30% value accretion if mortgage rates are down by another 50-100 bps.

China Property coverage valuation comparison

Exhibit 1: China developers' valuation comparisons

Company	Ticker	Daily liquidity (US\$ mn)		Type	Rating	Price as of 2/Jun/26	12 mth Price target	Potential upside/downside (%)	Target price disc. to NAV	End-26E NAV	Shr price (disc)/ prem to NAV	FD Core P/E (x)				P/B (x) (excl. revaluation gain)				Dividend yield (%)			
		Mkt Cap (US\$ bn)	1 mth trailing									25	26E	27E	28E	25	26E	27E	28E	25	26E	27E	28E
Developers																							
Stronger SOE developers																							
CMSK	001979.SZ	11.9	100	Central SOE	Neutral	8.94 (Rmb)	10.10	13	-15%	11.93	(25)	n.m.	22.1	18.4	16.9	0.8	0.8	0.8	0.8	0.6	2.0	2.5	2.7
COLI	0688.HK	22.7	95	Central SOE	Buy	16.3 (HK\$)	14.70	10	-10%	16.39	(1)	12.5	12.4	11.9	10.7	0.5	0.5	0.5	0.5	3.0	3.2	3.4	3.8
CR Land	1109.HK	32.9	154	Central SOE	Buy	36.2 (HK\$)	36.60	1	-10%	40.70	(11)	10.5	9.6	9.5	9.2	1.1	1.0	1.0	0.9	3.5	4.0	4.2	4.2
Greentown	3900.HK	2.9	15	Mixed ownership	Buy	8.9 (HK\$)	11.10	25	-15%	13.02	(32)	4.1	4.6	4.5	4.2	0.6	0.6	0.5	0.5	-	2.3	4.1	6.0
Jinmao	0817.HK	3.0	16	Central SOE	Buy	1.7 (HK\$)	1.70	0	-10%	1.89	(8)	44.3	24.7	12.9	7.0	0.7	0.7	0.7	0.6	1.7	1.7	3.3	6.1
Poly	600048.SS	9.9	136	Central SOE	Neutral	5.60 (Rmb)	6.50	16	-10%	7.24	(23)	64.8	n.m.	n.m.	n.m.	0.3	0.4	0.4	0.4	0.6	-	-	-
Average								7	-12%		(17)	12.5	12.4	11.9	9.2	0.7	0.7	0.6	0.6	1.6	2.2	2.9	3.8
POE developers																							
Gemdale	600383.SS	1.8	38	Mixed ownership	Sell	2.70 (Rmb)	2.70	0	-20%	3.40	(21)	n.m.	n.m.	n.m.	n.m.	0.3	0.4	0.4	0.4	-	-	-	-
Longfor	0960.HK	7.2	23	POE	Neutral	8.0 (HK\$)	8.70	9	-25%	11.55	(31)	n.m.	n.m.	23.8	15.6	0.4	0.4	0.4	0.4	1.0	-	1.3	2.1
Seazen	1030.HK	1.6	8	POE	Sell	1.8 (HK\$)	1.80	1	-50%	3.64	(51)	30.2	44.9	20.9	11.8	0.3	0.3	0.3	0.3	-	-	-	-
Average								3	-32%		(34)	30.2	44.9	22.3	13.7	0.4	0.4	0.4	0.4	0.3	-	0.4	0.7
Other SOE developers																							
OCT	000069.SZ	2.4	18	Central SOE	Sell	2.00 (Rmb)	1.50	25	-20%	n.m.	n.m.	n.m.	n.m.	n.m.	n.m.	0.4	0.5	0.5	0.6	-	-	-	-
Vanke (A)	000002.SZ	4.9	86	Mixed ownership	Sell	3.39 (Rmb)	3.70	9	-10%	4.10	(17)	n.m.	n.m.	n.m.	n.m.	0.4	0.4	0.5	0.5	-	-	-	-
Vanke (H)	2202.HK	0.8	11	Mixed ownership	Sell	2.78 (HK\$)	2.90	4	-35%	4.47	(38)	n.m.	n.m.	n.m.	n.m.	0.3	0.3	0.3	0.4	-	-	-	-
Average								(4)	-22%		(28)	n.m.	n.m.	n.m.	n.m.	0.3	0.4	0.4	0.5	-	-	-	-
Coverage average																							
H-share average																							
A-share average																							

Note:
Our 12-month target prices are based on end-2026E NAV for our coverage universe except for OCT.
Our OCT TP is based on SOTP of its property development (valued by NAV) and its tourism business (valued by P/E).

Source: Datastream, Goldman Sachs Global Investment Research

Disclosure Appendix

Reg AC

We, Yi Wang, CFA, Shi Xu and Kaiyan Jing, hereby certify that all of the views expressed in this report accurately reflect our personal views about the subject company or companies and its or their securities. We also certify that no part of our compensation was, is or will be, directly or indirectly, related to the specific recommendations or views expressed in this report.

Contributing Authors: Yi Wang, CFA Goldman Sachs (China) Securities Company Limited, Shi Xu Goldman Sachs (China) Securities Company Limited, Kaiyan Jing Goldman Sachs (China) Securities Company Limited.

Unless otherwise stated, the individuals listed in the Contributing Authors disclosure of this report are analysts in Goldman Sachs' Global Investment Research division.

GS Factor Profile

The Goldman Sachs Factor Profile provides investment context for a stock by comparing key attributes to the market (i.e. our universe of rated stocks) and its sector peers. The four key attributes depicted are: Growth, Financial Returns, Multiple (e.g. valuation) and Integrated (a composite of Growth, Financial Returns and Multiple). Growth, Financial Returns and Multiple are calculated by using normalized ranks for specific metrics for each stock. The normalized ranks for the metrics are then averaged and converted into percentiles for the relevant attribute. The precise calculation of each metric may vary depending on the fiscal year, industry and region, but the standard approach is as follows:

Growth is based on a stock's forward-looking sales growth, EBITDA growth and EPS growth (for financial stocks, only EPS and sales growth), with a higher percentile indicating a higher growth company. **Financial Returns** is based on a stock's forward-looking ROE, ROCE and CROCI (for financial stocks, only ROE), with a higher percentile indicating a company with higher financial returns. **Multiple** is based on a stock's forward-looking P/E, P/B, price/dividend (P/D), EV/EBITDA, EV/FCF and EV/Debt Adjusted Cash Flow (DACF) (for financial stocks, only P/E, P/B and P/D), with a higher percentile indicating a stock trading at a higher multiple. The **Integrated** percentile is calculated as the average of the Growth percentile, Financial Returns percentile and (100% - Multiple percentile).

Financial Returns and Multiple use the Goldman Sachs analyst forecasts at the fiscal year-end at least three quarters in the future. Growth uses inputs for the fiscal year at least seven quarters in the future compared with the year at least three quarters in the future (on a per-share basis for all metrics).

For a more detailed description of how we calculate the GS Factor Profile, please contact your GS representative.

M&A Rank

Across our global coverage, we examine stocks using an M&A framework, considering both qualitative factors and quantitative factors (which may vary across sectors and regions) to incorporate the potential that certain companies could be acquired. We then assign a M&A rank as a means of scoring companies under our rated coverage from 1 to 3, with 1 representing high (30%-50%) probability of the company becoming an acquisition target, 2 representing medium (15%-30%) probability and 3 representing low (0%-15%) probability. For companies ranked 1 or 2, in line with our standard departmental guidelines we incorporate an M&A component into our target price. M&A rank of 3 is considered immaterial and therefore does not factor into our price target, and may or may not be discussed in research.

Quantum

Quantum is Goldman Sachs' proprietary database providing access to detailed financial statement histories, forecasts and ratios. It can be used for in-depth analysis of a single company, or to make comparisons between companies in different sectors and markets.

Disclosures

Distribution of ratings/investment banking relationships

Goldman Sachs Investment Research global Equity coverage universe

	Rating Distribution				Investment Banking Relationships		
	Buy	Hold	Sell		Buy	Hold	Sell
Global	50%	34%	16%		65%	60%	45%

As of April 1, 2026, Goldman Sachs Global Investment Research had investment ratings on 3,074 equity securities. Goldman Sachs assigns stocks as Buys and Sells on various regional Investment Lists; stocks not so assigned are deemed Neutral. Such assignments equate to Buy, Hold and Sell for the purposes of the above disclosure required by the FINRA Rules. See 'Ratings, Coverage universe and related definitions' below. The Investment Banking Relationships chart reflects the percentage of subject companies within each rating category for whom Goldman Sachs has provided investment banking services within the previous twelve months.

Regulatory disclosures

Disclosures required by United States laws and regulations

See company-specific regulatory disclosures above for any of the following disclosures required as to companies referred to in this report: manager or co-manager in a pending transaction; 1% or other ownership; compensation for certain services; types of client relationships; managed/co-managed public offerings in prior periods; directorships; for equity securities, market making and/or specialist role. Goldman Sachs trades or may trade as a principal in debt securities (or in related derivatives) of issuers discussed in this report.

The following are additional required disclosures: **Ownership and material conflicts of interest:** Goldman Sachs policy prohibits its analysts, professionals reporting to analysts and members of their households from owning securities of any company in the analyst's area of coverage. **Analyst compensation:** Analysts are paid in part based on the profitability of Goldman Sachs, which includes investment banking revenues. **Analyst as officer or director:** Goldman Sachs policy generally prohibits its analysts, persons reporting to analysts or members of their households from serving as an officer, director or advisor of any company in the analyst's area of coverage. **Non-U.S. Analysts:** Non-U.S. analysts may not be associated persons of Goldman Sachs & Co. LLC and therefore may not be subject to FINRA Rule 2241 or FINRA Rule 2242 restrictions on communications with a subject company, public appearances and trading in securities covered by the analysts.

Distribution of ratings: See the distribution of ratings disclosure above. **Price chart:** See the price chart, with changes of ratings and price targets in prior periods, above, or, if electronic format or if with respect to multiple companies which are the subject of this report, on the Goldman Sachs website

at <https://www.gs.com/research/hedge.html>.

Additional disclosures required under the laws and regulations of jurisdictions other than the United States

The following disclosures are those required by the jurisdiction indicated, except to the extent already made above pursuant to United States laws and regulations. **Australia:** Goldman Sachs Australia Pty Ltd and its affiliates are not authorised deposit-taking institutions (as that term is defined in the Banking Act 1959 (Cth)) in Australia and do not provide banking services, nor carry on a banking business, in Australia. This research, and any access to it, is intended only for “wholesale clients” within the meaning of the Australian Corporations Act, unless otherwise agreed by Goldman Sachs. In producing research reports, members of Global Investment Research of Goldman Sachs Australia may attend site visits and other meetings hosted by the companies and other entities which are the subject of its research reports. In some instances the costs of such site visits or meetings may be met in part or in whole by the issuers concerned if Goldman Sachs Australia considers it is appropriate and reasonable in the specific circumstances relating to the site visit or meeting. To the extent that the contents of this document contains any financial product advice, it is general advice only and has been prepared by Goldman Sachs without taking into account a client’s objectives, financial situation or needs. A client should, before acting on any such advice, consider the appropriateness of the advice having regard to the client’s own objectives, financial situation and needs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests and a copy of Goldman Sachs’ Australian Sell-Side Research Independence Policy Statement are available at: <https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html>. **Brazil:** Disclosure information in relation to CVM Resolution n. 20 is available at <https://www.gs.com/worldwide/brazil/area/gir/index.html>. Where applicable, the Brazil-registered analyst primarily responsible for the content of this research report, as defined in Article 20 of CVM Resolution n. 20, is the first author named at the beginning of this report, unless indicated otherwise at the end of the text. **Canada:** This information is being provided to you for information purposes only and is not, and under no circumstances should be construed as, an advertisement, offering or solicitation by Goldman Sachs & Co. LLC for purchasers of securities in Canada to trade in any Canadian security. Goldman Sachs & Co. LLC is not registered as a dealer in any jurisdiction in Canada under applicable Canadian securities laws and generally is not permitted to trade in Canadian securities and may be prohibited from selling certain securities and products in certain jurisdictions in Canada. If you wish to trade in any Canadian securities or other products in Canada please contact Goldman Sachs Canada Inc., an affiliate of The Goldman Sachs Group Inc., or another registered Canadian dealer. **Hong Kong:** Further information on the securities of covered companies referred to in this research may be obtained on request from Goldman Sachs (Asia) L.L.C. **India:** Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (India) Securities Private Limited, Research Analyst - SEBI Registration Number INH000001493, 10th Floor, Ascent-Worli, Sudam Kalu Ahire Marg, Worli, Mumbai-400 025, India, Corporate Identity Number U74140MH2006FTC160634, Phone +91 22 6616 9000, Fax +91 22 6616 9001. Goldman Sachs may beneficially own 1% or more of the securities (as such term is defined in clause 2 (h) the Indian Securities Contracts (Regulation) Act, 1956) of the subject company or companies referred to in this research report. Investment in securities market are subject to market risks. Read all the related documents carefully before investing. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Goldman Sachs (India) Securities Private Limited compliance officer and investor grievance contact details can be found at: <https://www.goldmansachs.com/worldwide/india/documents/Grievance-Redressal-and-Escalation-Matrix.pdf>, and a copy of the annual audit compliance report can be found at this link: <https://publishing.gs.com/content/site/india-annual-compliance-report.html>. **Japan:** See below. **Korea:** This research, and any access to it, is intended only for “professional investors” within the meaning of the Financial Services and Capital Markets Act, unless otherwise agreed by Goldman Sachs. Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (Asia) L.L.C., Seoul Branch. **New Zealand:** Goldman Sachs New Zealand Limited and its affiliates are neither “registered banks” nor “deposit takers” (as defined in the Reserve Bank of New Zealand Act 1989) in New Zealand. This research, and any access to it, is intended for “wholesale clients” (as defined in the Financial Advisers Act 2008) unless otherwise agreed by Goldman Sachs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests is available at: <https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html>. **Russia:** Research reports distributed in the Russian Federation are not advertising as defined in the Russian legislation, but are information and analysis not having product promotion as their main purpose and do not provide appraisal within the meaning of the Russian legislation on appraisal activity. Research reports do not constitute a personalized investment recommendation as defined in Russian laws and regulations, are not addressed to a specific client, and are prepared without analyzing the financial circumstances, investment profiles or risk profiles of clients. Goldman Sachs assumes no responsibility for any investment decisions that may be taken by a client or any other person based on this research report. **Singapore:** Goldman Sachs (Singapore) Pte. (Company Number: 198602165W), which is regulated by the Monetary Authority of Singapore, accepts legal responsibility for this research, and should be contacted with respect to any matters arising from, or in connection with, this research. **Taiwan:** This material is for reference only and must not be reprinted without permission. Investors should carefully consider their own investment risk. Investment results are the responsibility of the individual investor. **United Kingdom:** Persons who would be categorized as retail clients in the United Kingdom, as such term is defined in the rules of the Financial Conduct Authority, should read this research in conjunction with prior Goldman Sachs research on the covered companies referred to herein and should refer to the risk warnings that have been sent to them by Goldman Sachs International. A copy of these risks warnings, and a glossary of certain financial terms used in this report, are available from Goldman Sachs International on request.

European Union and United Kingdom: Disclosure information in relation to Article 6 (2) of the European Commission Delegated Regulation (EU) (2016/958) supplementing Regulation (EU) No 596/2014 of the European Parliament and of the Council (including as that Delegated Regulation is implemented into United Kingdom domestic law and regulation following the United Kingdom’s departure from the European Union and the European Economic Area) with regard to regulatory technical standards for the technical arrangements for objective presentation of investment recommendations or other information recommending or suggesting an investment strategy and for disclosure of particular interests or indications of conflicts of interest is available at <https://www.gs.com/disclosures/europeanpolicy.html> which states the European Policy for Managing Conflicts of Interest in Connection with Investment Research.

Japan: Goldman Sachs Japan Co., Ltd. is a Financial Instrument Dealer registered with the Kanto Financial Bureau under registration number Kinsho 69, and a member of Japan Securities Dealers Association, Financial Futures Association of Japan Type II Financial Instruments Firms Association, and Investment Management Association of Japan. Sales and purchase of equities are subject to commission pre-determined with clients plus consumption tax. See company-specific disclosures as to any applicable disclosures required by Japanese stock exchanges, the Japanese Securities Dealers Association or the Japanese Securities Finance Company.

Ratings, coverage universe and related definitions

Buy (B), Neutral (N), Sell (S) Analysts recommend stocks as Buys or Sells for inclusion on various regional Investment Lists. Being assigned a Buy or Sell on an Investment List is determined by a stock’s total return potential relative to its coverage universe. Any stock not assigned as a Buy or a Sell on an Investment List with an active rating (i.e., a stock that is not Rating Suspended, Not Rated, Early-Stage Biotech, Coverage Suspended or Not Covered), is deemed Neutral. Each region manages Regional Conviction Lists, which are selected from Buy rated stocks on the respective region’s Investment Lists and represent investment recommendations focused on the size of the total return potential and/or the likelihood of the realization of the return across their respective areas of coverage. The addition or removal of stocks from such Conviction Lists are managed by the Investment Review Committee or other designated committee in each respective region and do not represent a change in the analysts’ investment rating for such stocks.

Total return potential represents the upside or downside differential between the current share price and the price target, including all paid or anticipated dividends, expected during the time horizon associated with the price target. Price targets are required for all covered stocks. The total return potential, price target and associated time horizon are stated in each report adding or reiterating an Investment List membership.

Coverage Universe: A list of all stocks in each coverage universe is available by primary analyst, stock and coverage universe at

<https://www.gs.com/research/hedge.html>.

Not Rated (NR). The investment rating, target price and earnings estimates (where relevant) are removed pursuant to Goldman Sachs policy when Goldman Sachs is acting in an advisory capacity in a merger or in a strategic transaction involving this company, when there are legal, regulatory or policy constraints due to Goldman Sachs' involvement in a transaction, and in certain other circumstances. **Early-Stage Biotech (ES).** An investment rating and a target price are not assigned pursuant to Goldman Sachs policy when this company has neither a drug, treatment or medical device that has passed a Phase II clinical trial nor a license to distribute a post-Phase II drug, treatment or medical device. **Rating Suspended (RS).** Goldman Sachs Research has suspended the investment rating and price target for this stock, because there is not a sufficient fundamental basis for determining an investment rating or target price. The previous investment rating and target price, if any, are no longer in effect for this stock and should not be relied upon. **Coverage Suspended (CS).** Goldman Sachs has suspended coverage of this company. **Not Covered (NC).** Goldman Sachs does not cover this company.

Global product; distributing entities

Goldman Sachs Global Investment Research produces and distributes research products for clients of Goldman Sachs on a global basis. Analysts based in Goldman Sachs offices around the world produce research on industries and companies, and research on macroeconomics, currencies, commodities and portfolio strategy. This research is disseminated in Australia by Goldman Sachs Australia Pty Ltd (ABN 21 006 797 897); in Brazil by Goldman Sachs do Brasil Corretora de Títulos e Valores Mobiliários S.A.; Public Communication Channel Goldman Sachs Brazil: 0800 727 5764 and / or contatogoldmanbrasil@gs.com. Available Weekdays (except holidays), from 9am to 6pm. Canal de Comunicação com o Público Goldman Sachs Brasil: 0800 727 5764 e/ou contatogoldmanbrasil@gs.com. Horário de funcionamento: segunda-feira à sexta-feira (exceto feriados), das 9h às 18h; in Canada by Goldman Sachs & Co. LLC; in Hong Kong by Goldman Sachs (Asia) L.L.C.; in India by Goldman Sachs (India) Securities Private Ltd.; in Japan by Goldman Sachs Japan Co., Ltd.; in the Republic of Korea by Goldman Sachs (Asia) L.L.C., Seoul Branch; in New Zealand by Goldman Sachs New Zealand Limited; in Russia by OOO Goldman Sachs; in Singapore by Goldman Sachs (Singapore) Pte. (Company Number: 198602165W); and in the United States of America by Goldman Sachs & Co. LLC. Goldman Sachs International has approved this research in connection with its distribution in the United Kingdom.

Goldman Sachs International ("GSI"), authorised by the Prudential Regulation Authority ("PRA") and regulated by the Financial Conduct Authority ("FCA") and the PRA, has approved this research in connection with its distribution in the United Kingdom.

European Economic Area: Goldman Sachs Bank Europe SE ("GSBE") is a credit institution incorporated in Germany and, within the Single Supervisory Mechanism, subject to direct prudential supervision by the European Central Bank and in other respects supervised by German Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, BaFin) and Deutsche Bundesbank and disseminates research within the European Economic Area.

General disclosures

This research is for our clients only. Other than disclosures relating to Goldman Sachs, this research is based on current public information that we consider reliable, but we do not represent it is accurate or complete, and it should not be relied on as such. The information, opinions, estimates and forecasts contained herein are as of the date hereof and are subject to change without prior notification. We seek to update our research as appropriate, but various regulations may prevent us from doing so. Other than certain industry reports published on a periodic basis, the large majority of reports are published at irregular intervals as appropriate in the analyst's judgment.

Goldman Sachs conducts a global full-service, integrated investment banking, investment management, and brokerage business. We have investment banking and other business relationships with a substantial percentage of the companies covered by Global Investment Research. Goldman Sachs & Co. LLC, the United States broker dealer, is a member of SIPC (<https://www.sipc.org>).

Our salespeople, traders, and other professionals may provide oral or written market commentary or trading strategies to our clients and principal trading desks that reflect opinions that are contrary to the opinions expressed in this research. Our asset management area, principal trading desks and investing businesses may make investment decisions that are inconsistent with the recommendations or views expressed in this research.

The analysts named in this report may have from time to time discussed with our clients, including Goldman Sachs salespersons and traders, or may discuss in this report, trading strategies that reference catalysts or events that may have a near-term impact on the market price of the equity securities discussed in this report, which impact may be directionally counter to the analyst's published price target expectations for such stocks. Any such trading strategies are distinct from and do not affect the analyst's fundamental equity rating for such stocks, which rating reflects a stock's return potential relative to its coverage universe as described herein.

We and our affiliates, officers, directors, and employees will from time to time have long or short positions in, act as principal in, and buy or sell, the securities or derivatives, if any, referred to in this research, unless otherwise prohibited by regulation or Goldman Sachs policy.

The views attributed to third party presenters at Goldman Sachs arranged conferences, including individuals from other parts of Goldman Sachs, do not necessarily reflect those of Global Investment Research and are not an official view of Goldman Sachs.

Any third party referenced herein, including any salespeople, traders and other professionals or members of their household, may have positions in the products mentioned that are inconsistent with the views expressed by analysts named in this report.

This research is not an offer to sell or the solicitation of an offer to buy any security in any jurisdiction where such an offer or solicitation would be illegal. It does not constitute a personal recommendation or take into account the particular investment objectives, financial situations, or needs of individual clients. Clients should consider whether any advice or recommendation in this research is suitable for their particular circumstances and, if appropriate, seek professional advice, including tax advice. The price and value of investments referred to in this research and the income from them may fluctuate. Past performance is not a guide to future performance, future returns are not guaranteed, and a loss of original capital may occur. Fluctuations in exchange rates could have adverse effects on the value or price of, or income derived from, certain investments.

Certain transactions, including those involving futures, options, and other derivatives, give rise to substantial risk and are not suitable for all investors. Investors should review current options and futures disclosure documents which are available from Goldman Sachs sales representatives or at <https://www.theocc.com/about/publications/character-risks.jsp> and https://www.goldmansachs.com/disclosures/cftc_fcm_disclosures. Transaction costs may be significant in option strategies calling for multiple purchase and sales of options such as spreads. Supporting documentation will be supplied upon request.

Differing Levels of Service provided by Global Investment Research: The level and types of services provided to you by Goldman Sachs Global Investment Research may vary as compared to that provided to internal and other external clients of GS, depending on various factors including your individual preferences as to the frequency and manner of receiving communication, your risk profile and investment focus and perspective (e.g., marketwide, sector specific, long term, short term), the size and scope of your overall client relationship with GS, and legal and regulatory constraints. As an example, certain clients may request to receive notifications when research on specific securities is published, and certain clients may request that specific data underlying analysts' fundamental analysis available on our internal client websites be delivered to them electronically through data feeds or otherwise. No change to an analyst's fundamental research views (e.g., ratings, price targets, or material changes to earnings estimates for

equity securities), will be communicated to any client prior to inclusion of such information in a research report broadly disseminated through electronic publication to our internal client websites or through other means, as necessary, to all clients who are entitled to receive such reports.

All research reports are disseminated and available to all clients simultaneously through electronic publication to our internal client websites. Not all research content is redistributed to our clients or available to third-party aggregators, nor is Goldman Sachs responsible for the redistribution of our research by third party aggregators. For research, models or other data related to one or more securities, markets or asset classes (including related services) that may be available to you, please contact your GS representative or go to <https://research.gs.com>.

Disclosure information is also available at <https://www.gs.com/research/hedge.html> or from Research Compliance, 200 West Street, New York, NY 10282.

© 2026 Goldman Sachs.

You are permitted to store, display, analyze, modify, reformat, and print the information made available to you via this service only for your own use. You may not resell or reverse engineer this information to calculate or develop any index for disclosure and/or marketing or create any other derivative works or commercial product(s), data or offering(s) without the express written consent of Goldman Sachs. You are not permitted to publish, transmit, or otherwise reproduce this information, in whole or in part, in any format to any third party without the express written consent of Goldman Sachs. This foregoing restriction includes, without limitation, using, extracting, downloading or retrieving this information, in whole or in part, to train or finetune a machine learning or artificial intelligence system, or to provide or reproduce this information, in whole or in part, as a prompt or input to any such system.